

Exercise Sheet 7: Properties of INNs

Due on 26.06.2026, 10:00
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Important Notes:

1. **Questions:** Please use the Moodle forum for questions. They will be answered by us or your fellow students.
2. **Deadline:** The due date for this exercise is **26.06.2026, 11:00**.
3. **Submission:** Submit one ZIP file containing your completed notebook and a single PDF report. Both files should contain your surname and matriculation number, e.g. *Surname-MatriculationNumber.zip*. The exported PDF should show your code, plots, and printed metrics. **Submissions that fail to follow the naming convention or miss the PDF/notebook files will not be graded.**

Use the notebook `exercise7.ipynb`. Complete all `# TODO` cells and keep the requested plots and printed metrics visible in your submitted notebook/PDF.

Task 1: Ambiguous Inverse Maps (7P)

Background. Invertible models are useful when a direct map loses information. Consider a point $x = (x_1, x_2) \in \mathbb{R}^2$ and an observation

$$y = \sqrt{x_1^2 + x_2^2}.$$

The map $x \mapsto y$ is not one-to-one: all points on the same circle have the same observation. A conditional invertible model can be viewed as storing the observed variable y together with a latent variable z , so that $x \leftrightarrow (y, z)$ becomes invertible.

Task.

1. (0.5P) Explain why a deterministic inverse from y to x cannot recover the original x .
2. (3P) Assume x is uniformly distributed on the unit circle, i.e. $y = 1$. Prove which deterministic prediction $\hat{x} \in \mathbb{R}^2$ minimizes the expected squared error $\mathbb{E}\|x - \hat{x}\|_2^2$.
3. (2P) Mathematically define a latent variable that stores the missing information for the circle example.
4. (1P) Mathematically define the complete forward and inverse maps between the point representation and the radius/latent representation.
5. (0.5P) State what information is stored in the latent variable in this example.

Task 2: Interpreting Autoencoder Latents (10P)

Background. This task is inspired by disentangling invertible interpretation networks, in particular the work of Esser et al. on explaining latent representations with an invertible map¹. An autoencoder provides a latent code z_{ae} for a colorized digit image. We then learn an invertible map

$$u = T(z_{ae}), \quad z_{ae} = T^{-1}(u),$$

where the transformed representation u is split into digit, color, and residual factors. The goal is not only reconstruction, but an interpretable coordinate system in latent space.

Task.

¹P. Esser, R. Rombach, and B. Ommer, *A Disentangling Invertible Interpretation Network for Explaining Latent Representations*, CVPR 2020.

1. (3P) Train the invertible interpretation network on the autoencoder latent codes.
2. (2P) Describe what information would be stored in the residual factor and why this factor is needed for the INN to work.
3. (5P) Use the inverse direction of the learned interpretation network for factor interventions. Produce one visualization where the digit factor is kept fixed while the color factor changes, and one visualization where the color factor is kept fixed while the digit factor changes.

Task 3: Conditional Flow on Colorized Digits (23P)

Background. Normalizing flows, such as RealNVP², are invertible neural networks that allow exact density evaluation through the change-of-variables formula

$$\log p_X(x) = \log p_Z(f(x)) + \log |\det J_f(x)|.$$

In this task, the flow is conditional: the digit label y is provided as additional information, and the model represents an invertible relation of the form

$$x \leftrightarrow z \mid y.$$

This allows both sampling in the inverse direction and likelihood evaluation under different digit conditions.

Task.

1. (6P) Train the conditional normalizing flow by completing the likelihood objective and training loop.
2. (6P) Evaluate each test image under all digit conditions and classify it by the condition with the best likelihood. Report the overall accuracy and the 10×10 confusion matrix.
3. (5P) Create a batch of random color-noise images, evaluate each image under all digit conditions, and compare the best likelihood scores of real digit images and random color noise using the requested side-by-side histograms.
4. (6P) Use the inverse direction of the conditional flow to inspect the roles of the condition and the latent variable. Generate samples for a fixed digit condition and transfer one encoded test image across all digit conditions.

²L. Dinh, J. Sohl-Dickstein, and S. Bengio, *Density Estimation using Real NVP*, 2016.